

Deploy an Automated Build Pipeline Using Docker Hub

Project structure

Program-6/

- app.py
- Requirements.txt
- Dockerfile

Step 1 : Create Github PAT

Github -> settings -> developer settings -> personal access tokens

choose classic tokens

generate with : write: package and read:package permissions

save token safely

Step 2 : Create docker file and build Image

docker build -t program6 .

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker build -t program-6 .
[+] Building 158.7s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 274B                                0.0s
=> [internal] load metadata for docker.io/library/node:20-alpine  17.1s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 2.69MB                                    0.1s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY package.json package-lock.json ./                  0.0s
=> [4/5] RUN npm install                                           141.3s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.0s
=> => writing image sha256:299d42db174382693e4049f098622ab4602bab54ae0d0 0.0s
=> => naming to docker.io/library/program-6                        0.0s
```

Step 3 : Tag image for Docker Hub

docker tag program6 -u <dockerhub_username>

Step 4 : Push image to docker hub

docker login -u <dockerhub-username>

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker tag program-6 darshanhegdep/program-6:latest
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker login -u darshanhegdep
```

 Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc031_darshan_hegde/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker push darshanhegdep/program-6:latest
The push refers to repository [docker.io/darshanhegdep/program-6]
73b5b2184ac9: Layer already exists
7c3d68c5aacb: Layer already exists
9cf5d14fc7b0: Layer already exists
eb1b0a5a5a32: Layer already exists
8bc61164599f: Layer already exists
a81608eb20af: Layer already exists
1548c3f692a1: Layer already exists
256f393e029f: Layer already exists
latest: digest: sha256:d4fe70b6edbdf02891ac90dadf66e0fa5fe175c4487df47d2950d84942cdaff7 size: 1990
```

Step 5: Tag image for github container registry (GHCR)

docker tag program6 ghcr.io/<github-username>/program6:latest

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker tag program-6 ghcr.io/darshanhegdep/program-6:latest
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ echo "ghp_HwcYqFLtVFWrPsHPvLhjG195a85o3d3wBwXf" | docker login ghcr.io -u darshanhegdep
WARNING! Your credentials are stored unencrypted in '/home/1rv24mc031_darshan_hegde/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/
Login Succeeded
```

Step 6 : Push Image to GHCR

Login using github pat

echo \$GITHUB_PAT | docker login ghcr.io -u <githubusername> --password-stdin

docker push ghcr.io/<githubusername>/program-6:latest

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ docker push ghcr.io/darshanhegdep/program-6:latest
The push refers to repository [ghcr.io/darshanhegdep/program-6]
73b5b2184ac9: Pushing 996.1kB/2.64MB
7c3d68c5aacb: Pushed
9cf5d14fc7b0: Pushed
eb1b0a5a5a32: Pushed
8bc61164599f: Pushed
a81608eb20af: Pushed
1548c3f692a1: Pushing 1.66MB/120.7MB
256f393e029f: Pushing 396.3kB/8.322MB
```

Step 7 : Connect Project to Github

git init

git remote add origin https://github.com/<github-username>/program-6.git

git remote -v

Step 8 : Push source code to github

git add .

git commit -m "Initial commit"

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git remote add origin https://github.com/DarshanHegdeP/Program-6.git
error: remote origin already exists.
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git remote set-url origin https://github.com/DarshanHegdeP/Program-6.git
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git remote -v
origin https://github.com/DarshanHegdeP/Program-6.git (fetch)
origin https://github.com/DarshanHegdeP/Program-6.git (push)
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git add .
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git commit -m "Initial"
[main 8599c0d] Initial
636 files changed, 72894 insertions(+)
create mode 100644 LabPrograms/Program6/Dockerfile
create mode 100644 LabPrograms/Program6/node_modules/.package-lock.json
create mode 100644 LabPrograms/Program6/node_modules/accepts/HISTORY.md
create mode 100644 LabPrograms/Program6/node_modules/accepts/LICENSE
create mode 100644 LabPrograms/Program6/node_modules/accepts/README.md
create mode 100644 LabPrograms/Program6/node_modules/accepts/index.js
```

git push -u origin main

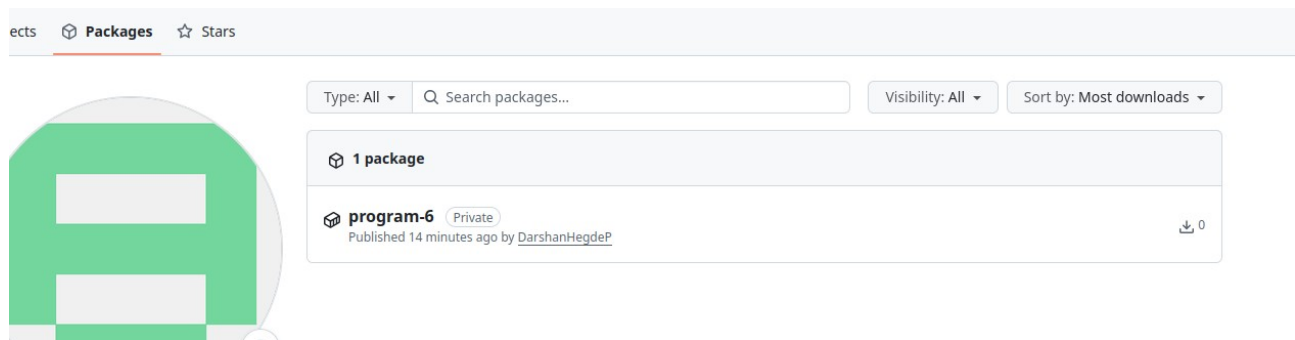
```
create mode 100644 LabPrograms/Program6/server.js
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/LabPrograms/Program6$ git push -u origin main
Username for 'https://github.com': DarshanHegdeP
Password for 'https://DarshanHegdeP@github.com':
Enumerating objects: 736, done.
Counting objects: 100% (736/736), done.
Delta compression using up to 4 threads
Compressing objects: 100% (539/539), done.
Writing objects: 100% (736/736), 725.31 KiB | 103.62 MiB/s, done.
Total 736 (delta 148), reused 690 (delta 138), pack-reused 0
remote: Resolving deltas: 100% (148/148), done.
To https://github.com/DarshanHegdeP/Program-6.git
 * [new branch]      main -> main
```

Step 9 : Verify Uploads

Github

verify github repo

check packages : image should appear as ghcr.io/<github_username>/program6 latest



Docker Hub

